

Qualification Pack



Heavy Earth Moving Machinery (HEMM) Electrician

Underground Metal/ Opencast/ Underground Coal

Electric Vehicle

QP Code: MIN/Q3102

Version: 2.0

NSQF Level: 4



Qualification Pack

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Qualification Pack

Contents

MIN/Q3102: Heavy Earth Moving Machinery (HEMM) Electrician	4
<i>Brief Job Description</i>	4
Applicable National Occupational Standards (NOS)	4
<i>Compulsory NOS</i>	4
<i>Elective 1: Underground Metal</i>	4
<i>Elective 2: Opencast</i>	4
<i>Elective 3: Underground Coal</i>	4
<i>Option: Electric Vehicle</i>	4
<i>Qualification Pack (QP) Parameters</i>	5
MIN/N3106: Carry out service, repair and overhaul of electrical and electronic aggregates of HEMM ...	7
DGT/VSQ/N0101: Employability Skills (30 Hours)	14
MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines	20
MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines	28
MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines	34
MIN/N3111: Conduct repair and maintenance of HEMM Electric Vehicle (EV)	43
Assessment Guidelines and Weightage	51
<i>Assessment Guidelines</i>	51
<i>Assessment Weightage</i>	52
Acronyms	54
Glossary	56

Qualification Pack

MIN/Q3102: Heavy Earth Moving Machinery (HEMM) Electrician

Brief Job Description

The individual in this role is responsible for the service, maintenance, repair and overhaul of electrical and electronic aggregates of HEMM available in mine.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it efficiently and effectively. The individual should be physically agile, strong, have good eye-sight and not suffer from color-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3106: Carry out service, repair and overhaul of electrical and electronic aggregates of HEMM](#)
2. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(*mandatory to select at least one*):

Elective 1: Underground Metal

1. [MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines](#)

Elective 2: Opencast

1. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)

Elective 3: Underground Coal

1. [MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines](#)

Options(*Not mandatory*):

Option : Electric Vehicle

Qualification Pack

1. [MIN/N3111: Conduct repair and maintenance of HEMM Electric Vehicle \(EV\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
Country	India
NSQF Level	4
Credits	22
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7412.0701
Minimum Educational Qualification & Experience	10th grade pass (2-year NTC /NAC or equivalent)(Mechanical/ Mining/Civil/ Electrical) OR Completed 2nd year of the 3-year diploma after 10 (Mechanical/ Mining/Civil/ Electrical) OR 12th grade Pass (or equivalent)(0.5 year relevant experience) OR 11th grade pass (1.5 year relevant experience) OR 10th grade pass (or equivalent)(3 years relevant experience) OR Previous relevant Qualification of NSQF Level (3)(3 years relevant experience)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Valid gas testing certificate (only for underground mines)
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	11/03/2029
NSQC Approval Date	11/03/2026



Skill Council For Mining Sector



MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP



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Version	2.0
Reference code on NQR	QG-04-Mining-05067-2026-V1-SCMS
NQR Version	1.0

Remarks:

Total Notional Duration: 450 Hours; Additional 90 hours for Optional NOS = Domain Theory Duration: 110 Hours; Additional 30 hours for Optional NOS Theory Duration+ Domain Practical Duration: 310 Hours; Additional 60 hours for Optional NOS Practical Duration+ Employability Skills Duration: 30 Hours

Qualification Pack

MIN/N3106: Carry out service, repair and overhaul of electrical and electronic aggregates of HEMM

Description

This OS unit is about carrying out diagnosis of fault, service and repairs of the electrical and electronic systems of a HEMM.

Scope

The scope covers the following :

- Prepare for service and repair work
- Perform service, repair and overhaul of electrical/electronic aggregates of HEMM
- Perform post service/repair activities

Elements and Performance Criteria

Prepare for service and repair work

To be competent, the user/individual on the job must be able to:

- PC1.** determine the work requirements such as type of tasks to be performed (repair, maintenance, etc.), HEMM to be maintained from work instructions, manuals, site visits, etc.
- PC2.** identify the auto component manufacturer specifications related to the various components/aggregates in the HEMM
- PC3.** review complaint sheet, log book and history sheet of the equipment and understand repair requirements
- PC4.** identify and arrange tools, consumables, spare parts, etc. required for performing service and repair work
- PC5.** inform relevant departments about the service and repair work to be performed and obtain approval before starting the work

Perform service, repair and overhaul of electrical/electronic aggregates of HEMM

To be competent, the user/individual on the job must be able to:

- PC6.** shut down the HEMM properly by disconnecting all primary and secondary energy sources safely
- PC7.** inspect the HEMM for defects and faulty parts/components by carrying out standard checks
- PC8.** diagnose direct or indirect faults in HEMM electrical/electronic systems as per OEM SOP
- PC9.** fix and verify that lockout has been successful and attach a lockout or tagout device to the HEMM to ensure that it cannot be started without removing the device
- PC10.** dismantle the HEMM and disconnect the faulty parts from it safely
- PC11.** check all wirings and electrical connections for any faults, damage, loose connections, etc.
- PC12.** inspect the HEMM for charge leakage, short circuit in parts, breakage of wires and clamps, unusual contact of electrical wires with moving parts, erratic/problematic performance or any other problem as reported in the complaint
- PC13.** repair all electrical and electronic faults including direct faults in input sensors, output actuators, wiring harnesses etc. as per SOP/WI

Qualification Pack

- PC14.** repair indirect faults in electrical/electronic aggregate due to another system/component
- PC15.** remove, replace, and repair relevant parts in various aggregate of the HEMM as per SOP
- PC16.** report any malfunctions/repairs in the machine beyond own scope to the concerned person

Perform post service/repair activities

To be competent, the user/individual on the job must be able to:

- PC17.** assemble back the covers, guards, clamps, insulation etc. of the HEMM after service and repair
- PC18.** update the Maintenance Data Sheet (MDS) on the HEMM
- PC19.** check the performance of HEMM post repair and report to supervisor/service advisor if further inspection is required by another specialist
- PC20.** record all repairs carried out, parts disposed and replaced, time taken and other significant findings observed during the work process
- PC21.** dispose-off materials such as old batteries, scrap of failed parts/aggregates as per organization's policies
- PC22.** report information such as service and work done, faults observed, action taken, parts replaced, etc. to the supervisors accurately

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** duties of workmen under the Mines Act-1952
- KU3.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU4.** different types of mines and detail of the mine one is working in
- KU5.** code of practice in specific areas of the mine
- KU6.** cost of equipment and loss for the company that results from damage of equipment
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to HEMM service and repair work
- KU8.** international safety guidelines and policies like COSHH, OSHA etc.
- KU9.** importance of safety in the vicinity of machinery
- KU10.** the outcome of violation of safety procedures
- KU11.** various types of HEMM i.e., excavators, backhoes, bulldozers, dump trucks, drill machines etc. utilized in mines
- KU12.** functioning of all HEMM and its various systems and components
- KU13.** normal running characteristics of all HEMM
- KU14.** machine specifications and elements of desired functioning of electrical equipment
- KU15.** tools and equipment required for carrying out repair and maintenance tasks
- KU16.** importance of obtaining approvals prior to starting the work
- KU17.** importance of maintenance schedules, work instructions and checklists recommended by the equipment manufacturer
- KU18.** significance of equipment layout and plant drawings for maintenance

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- KU19.** safe techniques for dismantling and assembling the electrical utilities
- KU20.** SOP recommended by the manufacturer for using tools, equipment, gauges, meters, testing equipment etc. during the service and repair activities
- KU21.** type of errors or defects in the tools/equipment
- KU22.** procedure to be followed for checking performance of various electrical and electronic components installed in HEMM
- KU23.** safety requirements to be followed during service and repair work
- KU24.** how to troubleshoot and perform minor repairs for electrical and electronic components installed
- KU25.** how to install basic electrical and electronic components in the various types of HEMM safely
- KU26.** faults and failures in engine, components/aggregates (including electrical and mechanical aggregates) and other units
- KU27.** corrective actions taken to address equipment faults
- KU28.** information to be recorded on the maintenance data sheet
- KU29.** documentation and records to be maintained regarding service and repair activities performed
- KU30.** how to ensure optimal operation of electrical and electronic components in HEMM
- KU31.** importance of proper disposal of failed components and changed oil, lubricant, grease etc. in accordance with safety, health and environmental policies and regulations such as battery, non-deployed airbag etc.
- KU32.** organisational and professional code of ethics and standards of practice
- KU33.** basic computer work
- KU34.** how to use services such as NEFT, IMPS, UPI, RTGS for online banking
- KU35.** importance of entrepreneurship and characteristics of an effective entrepreneur
- KU36.** various campaigns run by government nationally such as Swachh Bharat Abhiyan, make in India campaign etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** read and interpret symbols and readings
- GS3.** write work related information in English/regional language
- GS4.** communicate the work requirements to the supervisor and co-workers
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems

Qualification Pack

- GS12.** analyze and detect any potential problems which could arise during operation
- GS13.** report to the supervisor or deal with a colleague individually, depending on the type of concern

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for service and repair work</i>	5	10	-	5
PC1. determine the work requirements such as type of tasks to be performed (repair, maintenance, etc.), HEMM to be maintained from work instructions, manuals, site visits, etc.	1	2	-	1
PC2. identify the auto component manufacturer specifications related to the various components/aggregates in the HEMM	1	2	-	1
PC3. review complaint sheet, log book and history sheet of the equipment and understand repair requirements	1	2	-	1
PC4. identify and arrange tools, consumables, spare parts, etc. required for performing service and repair work	1	2	-	1
PC5. inform relevant departments about the service and repair work to be performed and obtain approval before starting the work	1	2	-	1
<i>Perform service, repair and overhaul of electrical/electronic aggregates of HEMM</i>	18	27	-	11
PC6. shut down the HEMM properly by disconnecting all primary and secondary energy sources safely	2	3	-	1
PC7. inspect the HEMM for defects and faulty parts/components by carrying out standard checks	2	3	-	1
PC8. diagnose direct or indirect faults in HEMM electrical/electronic systems as per OEM SOP	2	3	-	1
PC9. fix and verify that lockout has been successful and attach a lockout or tagout device to the HEMM to ensure that it cannot be started without removing the device	2	3	-	1
PC10. dismantle the HEMM and disconnect the faulty parts from it safely	2	3	-	1
PC11. check all wirings and electrical connections for any faults, damage, loose connections, etc.	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. inspect the HEMM for charge leakage, short circuit in parts, breakage of wires and clamps, unusual contact of electrical wires with moving parts, erratic/problematic performance or any other problem as reported in the complaint	2	2	-	1
PC13. repair all electrical and electronic faults including direct faults in input sensors, output actuators, wiring harnesses etc. as per SOP/WI	1	2	-	1
PC14. repair indirect faults in electrical/electronic aggregate due to another system/component	1	2	-	1
PC15. remove, replace, and repair relevant parts in various aggregate of the HEMM as per SOP	1	2	-	1
PC16. report any malfunctions/repairs in the machine beyond own scope to the concerned person	1	2	-	1
<i>Perform post service/repair activities</i>	7	13	-	4
PC17. assemble back the covers, guards, clamps, insulation etc. of the HEMM after service and repair	1	2	-	-
PC18. update the Maintenance Data Sheet (MDS) on the HEMM	1	2	-	1
PC19. check the performance of HEMM post repair and report to supervisor/service advisor if further inspection is required by another specialist	1	2	-	1
PC20. record all repairs carried out, parts disposed and replaced, time taken and other significant findings observed during the work process	1	2	-	1
PC21. dispose-off materials such as old batteries, scrap of failed parts/aggregates as per organization's policies	2	3	-	-
PC22. report information such as service and work done, faults observed, action taken, parts replaced, etc. to the supervisors accurately	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3106
NOS Name	Carry out service, repair and overhaul of electrical and electronic aggregates of HEMM
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4
Credits	12
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines (UMM) during the execution of various tasks and operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC2.** comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine
- PC3.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC4.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC5.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC6.** use self-rescue apparatus, appropriately when required
- PC7.** follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.
- PC8.** follow precautions against U/G electrical appliances
- PC9.** follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine
- PC10.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC11.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC12.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC13.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC14.** follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety
- PC15.** follow process for reporting any unsafe act/condition in work area to the concerned person

Qualification Pack

- PC16.** use underground mine communication system
- PC17.** ensure positive isolation near the work place if applicable
- PC18.** use appropriate PPE as per the requirement
- PC19.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC20.** disinfect the machine/tools before and after work/task
- PC21.** maintain hygiene at the work site
- PC22.** report any symptoms of illness to the shift-incharge
- PC23.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC24.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC25.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC26.** ensure that the stowing practices produce minimum disturbance to the surface
- PC27.** ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place
- PC28.** ensure the productivity of the machine for material/fuel conservation
- PC29.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC30.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases found in the mine and their effect
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (IME & PME) of person employed as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** importance of blind spot survey/training
- KU11.** about different types of machinery used in U/G mines
- KU12.** different types of supporting systems used in U/G mines as per SSP and SSR
- KU13.** about precautions to be taken when handling heavy equipment
- KU14.** various problems/incidents likely to occur
- KU15.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS

Qualification Pack

- KU16.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU17.** common sources of pollution in the mines and ways to minimize it
- KU18.** various types of fire extinguishers
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization as per DGMS guideline
- KU22.** mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU25.** rules and regulations for safety and security while handling hazardous materials
- KU26.** safety appliances and rescue equipment
- KU27.** importance and use of various communication system used in UG mines
- KU28.** importance of positive isolation at working site
- KU29.** safety and occupational health policy of organisation
- KU30.** six directional hazard identification process
- KU31.** basic personal and workplace hygiene
- KU32.** importance of FAB (Fresh Air Base)
- KU33.** basic provisions in Mines Creche Rules, 1966 (MCR) for females employed in the mines
- KU34.** the role and responsibilities of rescue room and rescue station and how to contact them in case of emergency
- KU35.** importance of taking shelter at the miner's station during blasting operation
- KU36.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU37.** importance of following infection control policies, '5-S' practices, and waste management
- KU38.** importance of water/material/energy conservation and management
- KU39.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from machine
- GS6.** use basic applications of a computer
- GS7.** plan and organize the work order and tasks
- GS8.** use reasoning skills to identify and resolve fundamental problems

Qualification Pack

GS9. complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	42	-	17
PC1. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	-	2	-	2
PC2. comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine	1	3	-	-
PC3. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	4	-	1
PC4. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	1	1	-	-
PC5. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	1	2	-	1
PC6. use self-rescue apparatus, appropriately when required	-	4	-	1
PC7. follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.	1	2	-	1
PC8. follow precautions against U/G electrical appliances	2	2	-	2
PC9. follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine	2	2	-	1
PC10. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	2	4	-	-
PC11. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	1	1	-	-
PC13. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	-	-	-
PC14. follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety	1	2	-	1
PC15. follow process for reporting any unsafe act/condition in work area to the concerned person	-	2	-	1
PC16. use underground mine communication system	-	1	-	-
PC17. ensure positive isolation near the work place if applicable	1	1	-	1
PC18. use appropriate PPE as per the requirement	2	4	-	2
PC19. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC20. disinfect the machine/tools before and after work/task	1	1	-	-
PC21. maintain hygiene at the work site	1	1	-	1
PC22. report any symptoms of illness to the shift-incharge	1	-	-	-
PC23. identify six directional hazards at workplace and take decisions accordingly	2	-	-	-
<i>Follow environmental guidelines</i>	6	8	-	3
PC24. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	2	-	-
PC25. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	-	1	-	-
PC26. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place	1	-	-	1
PC28. ensure the productivity of the machine for material/fuel conservation	1	1	-	-
PC29. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	1	-	1
PC30. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1702
NOS Name	Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

Qualification Pack

- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



Skill Council For Mining Sector



Qualification Pack

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** follow preventive measures against firedamp, whitedamp, blackdamp etc.
- PC2.** use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)
- PC3.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC4.** comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine
- PC5.** ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle
- PC6.** ensure that every instrument, apparatus and equipment are DGMS approved before these are used
- PC7.** ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining
- PC8.** provide first aid to an injured person
- PC9.** follow safety precautions against spontaneous heating of the coal
- PC10.** operate various types of fire extinguishers to control different types of fire at worksite, if required
- PC11.** ensure that no person is traveling/working/staying under unsupported roof
- PC12.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC13.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC14.** use self-rescue apparatus appropriately when required
- PC15.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC16.** follow precautions against U/G electrical appliances

Qualification Pack

- PC17.** take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports
- PC18.** follow appropriate SOP while working near any isolated and sealed off area of the mine
- PC19.** ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly
- PC20.** take relevant safety precautions during depillaring operation in UCM
- PC21.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC22.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC23.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC24.** follow laid out SOP in case of alarm signal for leakage of inflammable gases
- PC25.** follow the process of reporting any unsafe act/condition in the working area to the concerned person
- PC26.** use underground mine communication system
- PC27.** ensure positive isolation near the work place if applicable
- PC28.** use appropriate Personal Protective Equipment (PPE) as per the requirement
- PC29.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC30.** disinfect the machine/tools before and after work/task
- PC31.** maintain hygiene at the work site
- PC32.** report any symptoms of illness to the shift-incharge
- PC33.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC34.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC35.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC36.** ensure that the stowing practices produce minimum disturbance to the surface
- PC37.** ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place
- PC38.** ensure the productivity of the machine for material/fuel conservation
- PC39.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC40.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)

Qualification Pack

- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases available in the mine and their effects; and their control measures
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (Initial Medical Examination (IME) & Periodical Medical Examination (PME)) of a person employed, as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** about different types of machineries used in U/G mines
- KU11.** different types of supporting system used in U/G mines as per SSP and SSR
- KU12.** about precautions to be taken when handling heavy equipment
- KU13.** various problems/incidents likely to occur
- KU14.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS
- KU15.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU16.** common sources of pollution in the mines and ways to minimize it
- KU17.** various types of fire extinguishers
- KU18.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU19.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU20.** mining area-specific signs, and other safety and emergency signals
- KU21.** the outcome of violation of safety procedures
- KU22.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU23.** hazardous material safety, security rules and regulations
- KU24.** safety appliances and rescue equipment
- KU25.** importance and use of various communication system used in UG mines
- KU26.** importance of positive isolation at working site
- KU27.** safety and occupational health policy of organisation
- KU28.** six directional hazard identification process
- KU29.** importance of blind spot survey/training
- KU30.** basic personal and workplace hygiene
- KU31.** importance of FAB (Fresh Air Base)
- KU32.** basic provisions in Mines Creche Rules, 1966 (MCR) for any females employed in the mines
- KU33.** about basic safety regulations of Coal Mines Regulation, 2017 (CMR)
- KU34.** types of stone dust barrier and its importance
- KU35.** coal dust explosion and its preventive measures
- KU36.** classification of coal mines as per the degree of gassiness of coal seams such as first degree, second degree, and third-degree mines
- KU37.** precautions as per the gassiness of the coal mines

Qualification Pack

- KU38.** use of flame safety lamp and its parts
- KU39.** about coal mines occupational disease such as pneumoconiosis or 'black lung' and their preventive measures
- KU40.** Standard of Ventilation as per the section 153 of the CMR 2017
- KU41.** Standard of Lighting as per the section 178 of the CMR 2017
- KU42.** the roles, duties and responsibilities of rescue team members, rescue room and rescue station and how to contact them in case of emergency
- KU43.** the correct steps for conducting any rescue work as per Mine Rescue Rule (MRR)
- KU44.** importance of taking shelter at the miner's station during blasting operation
- KU45.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU46.** importance of following infection control policies, '5-S' practices, and waste management
- KU47.** importance of water/material/energy conservation and management
- KU48.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from a machine
- GS6.** plan and organize the work order and tasks
- GS7.** use basic applications of the computer
- GS8.** use reasoning skills to identify and resolve fundamental problems
- GS9.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	43	-	14
PC1. follow preventive measures against firedamp, whitedamp, blackdamp etc.	1	1	-	-
PC2. use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)	-	1	-	-
PC3. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	1	1	-	1
PC4. comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine	1	1	-	1
PC5. ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle	-	1	-	-
PC6. ensure that every instrument, apparatus and equipment are DGMS approved before these are used	1	1	-	-
PC7. ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining	3	3	-	-
PC8. provide first aid to an injured person	1	2	-	1
PC9. follow safety precautions against spontaneous heating of the coal	1	1	-	-
PC10. operate various types of fire extinguishers to control different types of fire at worksite, if required	-	2	-	-
PC11. ensure that no person is traveling/working/staying under unsupported roof	1	-	-	-
PC12. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	-	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	-	1	-	1
PC14. use self-rescue apparatus appropriately when required	-	2	-	-
PC15. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC16. follow precautions against U/G electrical appliances	2	2	-	1
PC17. take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports	1	2	-	-
PC18. follow appropriate SOP while working near any isolated and sealed off area of the mine	1	1	-	1
PC19. ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly	1	1	-	-
PC20. take relevant safety precautions during depillaring operation in UCM	-	2	-	1
PC21. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	1	1	-	-
PC22. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	-	1	-	1
PC23. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	-	1	-	1
PC24. follow laid out SOP in case of alarm signal for leakage of inflammable gases	1	1	-	-
PC25. follow the process of reporting any unsafe act/condition in the working area to the concerned person	-	1	-	1
PC26. use underground mine communication system	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure positive isolation near the work place if applicable	1	1	-	-
PC28. use appropriate Personal Protective Equipment (PPE) as per the requirement	2	2	-	1
PC29. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	-	1	-	-
PC30. disinfect the machine/tools before and after work/task	1	1	-	-
PC31. maintain hygiene at the work site	1	1	-	-
PC32. report any symptoms of illness to the shift-incharge	1	1	-	-
PC33. identify six directional hazards at workplace and take decisions accordingly	-	1	-	1
<i>Follow environmental guidelines</i>	6	7	-	6
PC34. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	1	-	1
PC35. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	1	1	-	1
PC36. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-
PC37. ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place	1	1	-	1
PC38. ensure the productivity of the machine for material/fuel conservation	1	1	-	1
PC39. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	-	1	-	1
PC40. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1704
NOS Name	Follow Health, Safety, and Environmental guidelines for underground coal mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

MIN/N3111: Conduct repair and maintenance of HEMM Electric Vehicle (EV)

Description

This unit is about the repair and maintenance of Heavy Earth Moving Machinery (HEMM) Electric Vehicles (EVs). It includes routine inspections, diagnosing faults, and performing repairs on electrical and mechanical components to ensure optimal performance, safety, and longevity of the vehicle.

Scope

The scope covers the following :

- Maintain electrical systems
- Maintain battery and motors
- Maintain and repair cooling systems
- Maintain charging infrastructure

Elements and Performance Criteria

Maintain electrical systems

To be competent, the user/individual on the job must be able to:

- PC1.** conduct a visual and physical inspection of the electrical input and output bus for signs of corrosion, looseness, or damage
- PC2.** use thermal imaging or infrared cameras to detect overheating or loose connections
- PC3.** check for damaged insulation and shielding to prevent electrical short circuits or exposure risks
- PC4.** verify that bus bars and connectors are securely fastened and free from external contaminants such as dirt, moisture, or oil
- PC5.** utilize specialized diagnostic tools such as multimeters, clamp meters, and oscilloscopes to measure voltage, current, and resistance of the electrical system
- PC6.** identify and address irregularities in current flow that could indicate underlying faults or inefficiencies
- PC7.** inspect and clean sensors (e.g., motor temperature sensors, battery cell voltage sensors, high-voltage insulation monitoring sensors, accelerator pedal position sensors, brake pressure sensors, inverter temperature sensors, etc.) for dust, grease, or contamination that may interfere with accurate readings
- PC8.** use Electronic Control Unit (ECU) tools to verify sensor functionality, ensuring they provide correct feedback on battery health, motor performance, and system temperatures
- PC9.** test solenoids for proper actuation, responsiveness, and electrical continuity
- PC10.** replace faulty sensors or solenoids with Original Equipment Manufacturer (OEM) -approved components to maintain system reliability
- PC11.** conduct a thorough inspection of the wiring harness for signs of chafing, cuts, and exposure to environmental hazards such as water or extreme temperatures

Qualification Pack

- PC12.** check for loose, corroded or damaged connectors that could result in poor signal transmission
- PC13.** repair or replace damaged wiring harness sections using OEM-approved components and follow correct crimping and splicing procedures
- PC14.** use appropriate lockout/tagout (LOTO) procedures to prevent accidental energization during maintenance

Maintain battery and motors

To be competent, the user/individual on the job must be able to:

- PC15.** regularly inspect battery state of charge (SOC) to ensure optimal performance
- PC16.** measure and analyze battery voltage, current, and temperature using diagnostic tools
- PC17.** identify and address signs of swelling, leakage, or corrosion on battery modules
- PC18.** perform battery equalization to balance charge levels across all cells (cell balancing)
- PC19.** ensure battery connectors are secure and free from dirt deposits, moisture, or corrosion
- PC20.** verify proper connection of charging cables and inspect for wear or damage
- PC21.** calibrate and update Battery Management System (BMS) software for precise battery performance tracking
- PC22.** check motor bearings for wear, lubrication levels, and smooth rotation
- PC23.** use thermal imaging or sensors to detect overheating issues in the motor
- PC24.** monitor electric motor efficiency, torque output, and vibration levels
- PC25.** check gearbox oil levels and ensure oil quality meets manufacturer specifications
- PC26.** inspect seals and gaskets for leaks and replace them if necessary
- PC27.** monitor motor and gearbox temperatures to prevent thermal damage
- PC28.** verify fan operation and replace damaged components to ensure effective cooling

Maintain and repair cooling systems

To be competent, the user/individual on the job must be able to:

- PC29.** maintain proper coolant levels and check for potential leaks in cooling lines
- PC30.** inspect hoses, pipes, and connectors for leaks, cracks, or deterioration
- PC31.** test the thermostat for proper operation and replace it, if it is faulty
- PC32.** monitor battery and motor temperatures during operation to ensure efficient thermal management
- PC33.** inspect the cooling fan for smooth operation, ensuring it engages at the correct temperature
- PC34.** check belt tension and alignment, adjusting or replacing worn-out belts to prevent slippage
- PC35.** test the hydraulic pump for efficiency and listen for abnormal noises indicating potential failure
- PC36.** regularly check hydraulic fluid levels, top up with the appropriate fluid grade, and periodically flush and replace the fluid to prevent contamination and maintain optimal operation
- PC37.** use diagnostic software to monitor temperature fluctuations in key EV components
- PC38.** replace worn-out or malfunctioning thermostats, sensors, or cooling system components as per OEM guidelines
- PC39.** ensure proper circulation of coolant through all required components, checking for any blockages

Qualification Pack

Maintain charging infrastructure

To be competent, the user/individual on the job must be able to:

- PC40.** conduct routine inspections of charging stations to ensure they are operational and free from physical damage
- PC41.** check for any signs of wear and tear, corrosion, or loose connections in charging equipment
- PC42.** verify that all charging station components, including power supply units, cooling fans, and safety features, are functioning properly
- PC43.** examine charging cables for any visible damage, frayed wires, or damaged insulation that could lead to electrical hazards
- PC44.** ensure connectors are free of dirt, dust, or moisture that may interfere with proper connection and power transfer
- PC45.** replace any worn-out or damaged charging cables and connectors following OEM specifications
- PC46.** use multimeters and diagnostic tools to measure output voltage, current, and power levels to confirm they meet manufacturer specifications
- PC47.** perform load testing on charging stations to ensure consistent and stable power delivery
- PC48.** identify common charging station malfunctions, including power fluctuations, overheating, and communication errors with vehicles

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** working principles of high-voltage electrical input and output bus systems in Heavy Earth Moving Machinery (HEMM) EVs
- KU2.** importance of insulation, shielding, and secure electrical connections to prevent short circuits and exposure risks
- KU3.** role of sensors and solenoids in monitoring and controlling EV systems, including battery health, motor performance, and temperature regulation
- KU4.** structure and function of wiring harnesses, including the identification of high-voltage and low-voltage circuits
- KU5.** signs of battery degradation, including swelling, leakage, and corrosion, and their impact on EV performance
- KU6.** function and calibration of the Battery Management System (BMS) for optimal battery performance tracking
- KU7.** gearbox oil specifications and the importance of maintaining proper oil levels to prevent transmission damage
- KU8.** importance of thermal management in EVs, including cooling requirements for batteries, motors, and power electronics
- KU9.** structure and operation of EV cooling systems, including radiators, cooling fans, thermostats, and heat exchangers
- KU10.** role of hydraulic systems in HEMM EVs, including fluid maintenance, pump efficiency, and contamination prevention
- KU11.** operational principles of EV charging stations, including power supply management and energy conversion

Qualification Pack

- KU12.** effects of corrosion, dirt, and moisture on charging connectors and how to mitigate these risks
- KU13.** methods for measuring voltage, current, and power output at charging stations to verify proper functionality
- KU14.** load testing techniques for ensuring stable and consistent power delivery to EVs
- KU15.** significance of electrical grounding in charging infrastructure and methods for testing earthing resistance levels
- KU16.** common charging station malfunctions, such as overheating, power fluctuations, and communication failures, and troubleshooting techniques
- KU17.** electrical safety measures to prevent accidental shocks, fires, or power surges during charging station maintenance
- KU18.** safety procedures and personal protective equipment (PPE) requirements when handling high-voltage components in HEMM EVs
- KU19.** industry standards and regulations for EV maintenance, including environmental and workplace safety compliance
- KU20.** emergency response protocols for dealing with electrical faults, battery fires, or system failures
- KU21.** importance of adhering to OEM guidelines and manufacturer specifications for all maintenance and repair work
- KU22.** role of predictive and preventive maintenance in extending the lifespan and efficiency of EV components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** read and interpret symbols and readings
- GS3.** write work related information in English/regional language
- GS4.** communicate the work requirements to the supervisor and co-workers
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems
- GS12.** analyze and detect any potential problems which could arise during operation
- GS13.** report to the supervisor or deal with a colleague individually, depending on the type of concern

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain electrical systems</i>	7	13	-	5
PC1. conduct a visual and physical inspection of the electrical input and output bus for signs of corrosion, looseness, or damage	-	1	-	1
PC2. use thermal imaging or infrared cameras to detect overheating or loose connections	-	1	-	1
PC3. check for damaged insulation and shielding to prevent electrical short circuits or exposure risks	1	1	-	-
PC4. verify that bus bars and connectors are securely fastened and free from external contaminants such as dirt, moisture, or oil	1	1	-	-
PC5. utilize specialized diagnostic tools such as multimeters, clamp meters, and oscilloscopes to measure voltage, current, and resistance of the electrical system	1	1	-	-
PC6. identify and address irregularities in current flow that could indicate underlying faults or inefficiencies	-	1	-	-
PC7. inspect and clean sensors (e.g., motor temperature sensors, battery cell voltage sensors, high-voltage insulation monitoring sensors, accelerator pedal position sensors, brake pressure sensors, inverter temperature sensors, etc.) for dust, grease, or contamination that may interfere with accurate readings	-	1	-	-
PC8. use Electronic Control Unit (ECU) tools to verify sensor functionality, ensuring they provide correct feedback on battery health, motor performance, and system temperatures	1	1	-	1
PC9. test solenoids for proper actuation, responsiveness, and electrical continuity	1	1	-	-
PC10. replace faulty sensors or solenoids with Original Equipment Manufacturer (OEM) -approved components to maintain system reliability	1	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. conduct a thorough inspection of the wiring harness for signs of chafing, cuts, and exposure to environmental hazards such as water or extreme temperatures	1	-	-	-
PC12. check for loose, corroded or damaged connectors that could result in poor signal transmission	-	1	-	1
PC13. repair or replace damaged wiring harness sections using OEM-approved components and follow correct crimping and splicing procedures	-	1	-	-
PC14. use appropriate lockout/tagout (LOTO) procedures to prevent accidental energization during maintenance	-	1	-	-
<i>Maintain battery and motors</i>	8	12	-	4
PC15. regularly inspect battery state of charge (SOC) to ensure optimal performance	1	1	-	-
PC16. measure and analyze battery voltage, current, and temperature using diagnostic tools	1	1	-	-
PC17. identify and address signs of swelling, leakage, or corrosion on battery modules	-	1	-	-
PC18. perform battery equalization to balance charge levels across all cells (cell balancing)	-	1	-	-
PC19. ensure battery connectors are secure and free from dirt deposits, moisture, or corrosion	-	1	-	-
PC20. verify proper connection of charging cables and inspect for wear or damage	-	1	-	1
PC21. calibrate and update Battery Management System (BMS) software for precise battery performance tracking	-	1	-	1
PC22. check motor bearings for wear, lubrication levels, and smooth rotation	1	1	-	-
PC23. use thermal imaging or sensors to detect overheating issues in the motor	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. monitor electric motor efficiency, torque output, and vibration levels	1	1	-	-
PC25. check gearbox oil levels and ensure oil quality meets manufacturer specifications	1	-	-	-
PC26. inspect seals and gaskets for leaks and replace them if necessary	1	-	-	-
PC27. monitor motor and gearbox temperatures to prevent thermal damage	1	1	-	1
PC28. verify fan operation and replace damaged components to ensure effective cooling	-	1	-	1
<i>Maintain and repair cooling systems</i>	7	15	-	5
PC29. maintain proper coolant levels and check for potential leaks in cooling lines	1	1	-	-
PC30. inspect hoses, pipes, and connectors for leaks, cracks, or deterioration	-	1	-	-
PC31. test the thermostat for proper operation and replace it, if it is faulty	-	1	-	-
PC32. monitor battery and motor temperatures during operation to ensure efficient thermal management	1	1	-	1
PC33. inspect the cooling fan for smooth operation, ensuring it engages at the correct temperature	1	1	-	1
PC34. check belt tension and alignment, adjusting or replacing worn-out belts to prevent slippage	-	2	-	1
PC35. test the hydraulic pump for efficiency and listen for abnormal noises indicating potential failure	-	1	-	-
PC36. regularly check hydraulic fluid levels, top up with the appropriate fluid grade, and periodically flush and replace the fluid to prevent contamination and maintain optimal operation	1	1	-	-
PC37. use diagnostic software to monitor temperature fluctuations in key EV components	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC38. replace worn-out or malfunctioning thermostats, sensors, or cooling system components as per OEM guidelines	1	2	-	1
PC39. ensure proper circulation of coolant through all required components, checking for any blockages	1	2	-	-
<i>Maintain charging infrastructure</i>	8	10	-	6
PC40. conduct routine inspections of charging stations to ensure they are operational and free from physical damage	1	1	-	-
PC41. check for any signs of wear and tear, corrosion, or loose connections in charging equipment	1	1	-	-
PC42. verify that all charging station components, including power supply units, cooling fans, and safety features, are functioning properly	1	2	-	1
PC43. examine charging cables for any visible damage, frayed wires, or damaged insulation that could lead to electrical hazards	1	1	-	1
PC44. ensure connectors are free of dirt, dust, or moisture that may interfere with proper connection and power transfer	-	1	-	1
PC45. replace any worn-out or damaged charging cables and connectors following OEM specifications	1	1	-	1
PC46. use multimeters and diagnostic tools to measure output voltage, current, and power levels to confirm they meet manufacturer specifications	1	1	-	1
PC47. perform load testing on charging stations to ensure consistent and stable power delivery	1	1	-	1
PC48. identify common charging station malfunctions, including power fluctuations, overheating, and communication errors with vehicles	1	1	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3111
NOS Name	Conduct repair and maintenance of HEMM Electric Vehicle (EV)
Sector	Mining
Sub-Sector	
Occupation	Electrical Services
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3106.Carry out service, repair and overhaul of electrical and electronic aggregates of HEMM	30	50	0	20	100	40
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	20
Total	50	80	-	20	150	60

Elective: 1 Underground Metal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1702.Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines	30	50	-	20	100	40
Total	30	50	-	20	100	40

Elective: 2 Opencast

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	-	20	100	40
Total	30	50	-	20	100	40

Elective: 3 Underground Coal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1704.Follow Health, Safety, and Environmental guidelines for underground coal mines	30	50	-	20	100	40
Total	30	50	-	20	100	40

Optional: 1 Electric Vehicle

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3111.Conduct repair and maintenance of HEMM Electric Vehicle (EV)	30	50	0	20	100	40
Total	30	50	-	20	100	40

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
QP	Qualification Pack
NSQF	National Skills Qualifications Framework
TVET	Technical and Vocational Education and Training
SCMS	Skill Council for Mining Sector
ROPS	Roll-Over Protection Structure
FOPS	Falling Object Protective Structure
PC	Performance Criteria
HEMM	Heavy Earth Moving Machinery
OEM	Original Equipment Manufacturer
DGMS	Directorate General of Mining Safety
MMR	Metalliferous Mines Regulations
CMR	Coal Mines Regulations
SOP	Standard Operating Practice
MVTR	Mine Vocational Training Rules
MRR	Mines Rescue Rules
HSE	health, safety & environment
OB	Over Burden
ERP	Enterprise resource planning
MCDR	Mineral Conservation and Development Rules
NOS	National Occupational Standard(s)

Qualification Pack

QP	Qualification Pack
NSQF	National Skills Qualifications Framework
TVET	Technical and Vocational Education and Training
SCMS	Skill Council for Mining Sector
ROPS	Roll-Over Protection Structure
FOPS	Falling Object Protective Structure
PC	Performance Criteria
HEMM	Heavy Earth Moving Machinery
OEM	Original Equipment Manufacturer
DGMS	Directorate General of Mining Safety
MMR	Metalliferous Mines Regulations
CMR	Coal Mines Regulations
SOP	Standard Operating Practice
MVTR	Mine Vocational Training Rules
MRR	Mines Rescue Rules
HSE	health, safety & environment
OB	Over Burden
ERP	Enterprise resource planning
MCDR	Mineral Conservation and Development Rules

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	A sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
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Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.

Qualification Pack

Qualification Pack	QP comprises the set of OS, together with the educational, training, and other criteria required to perform a job role. A QP is assigned a unique qualifications pack role
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Sector	A sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
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